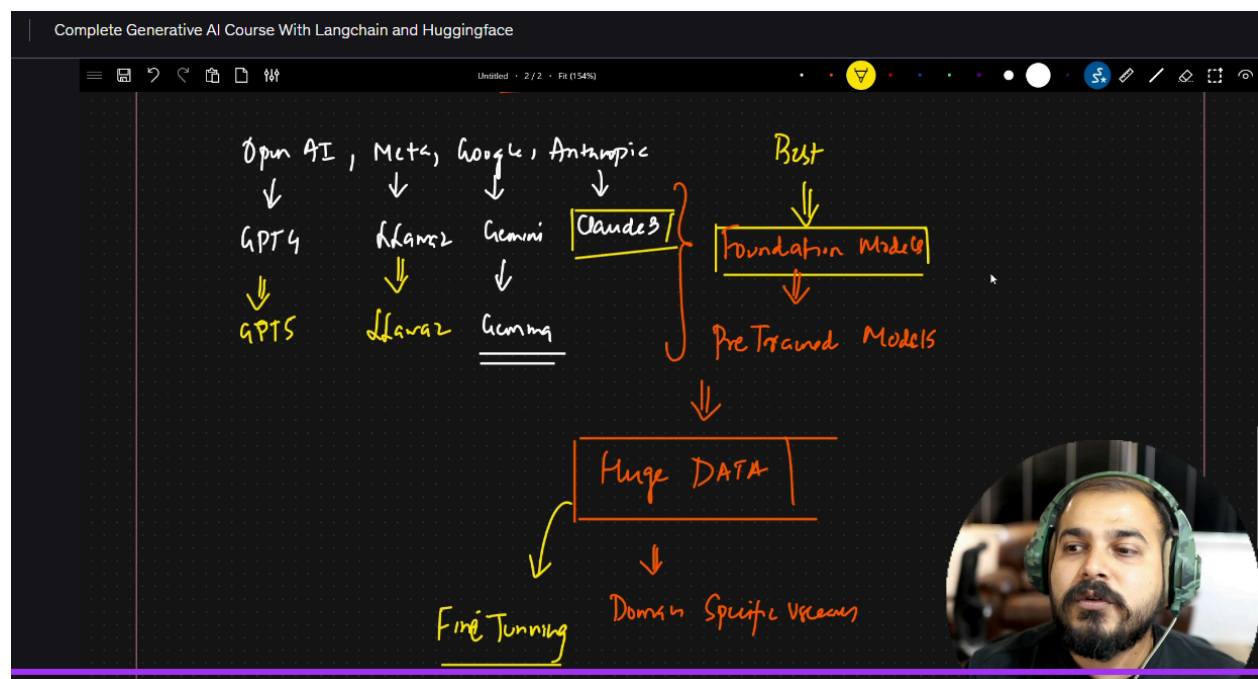
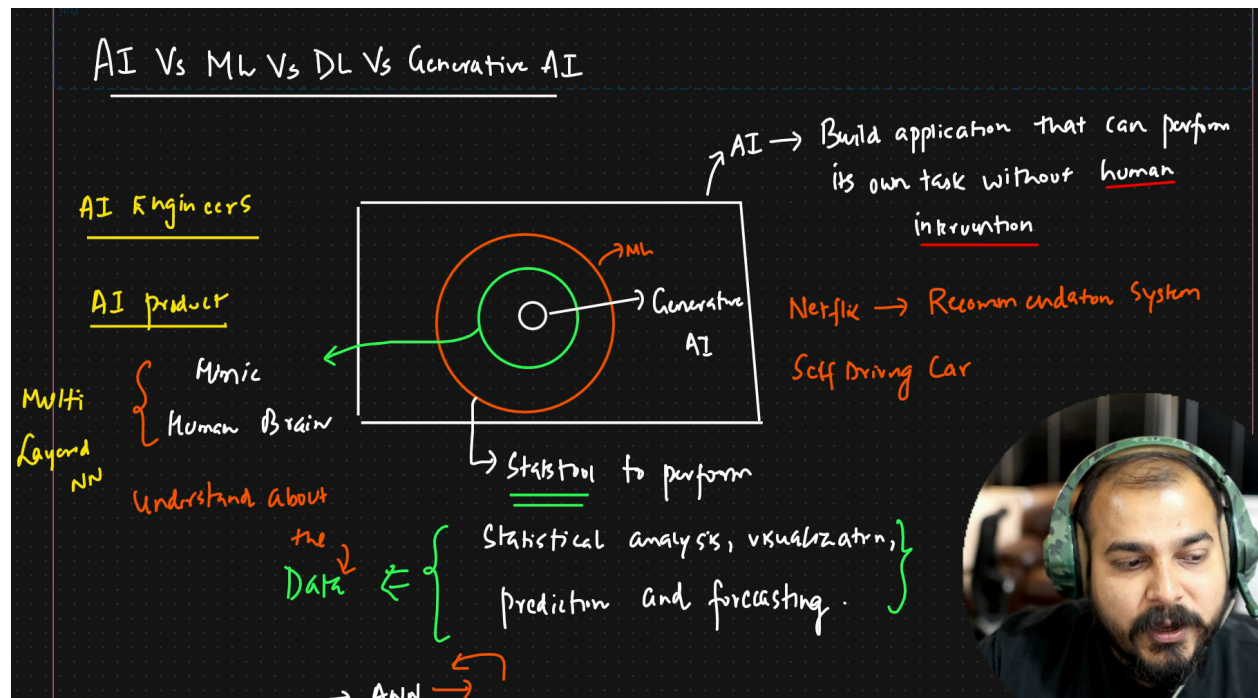
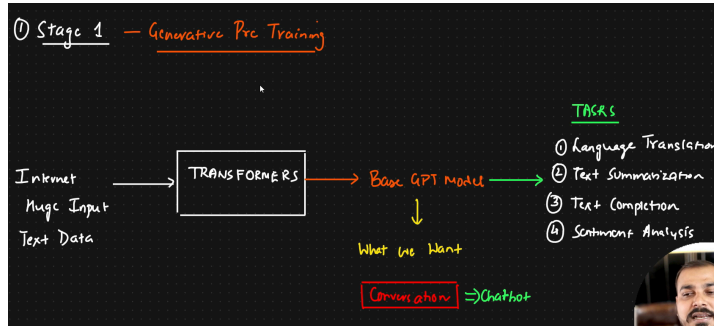
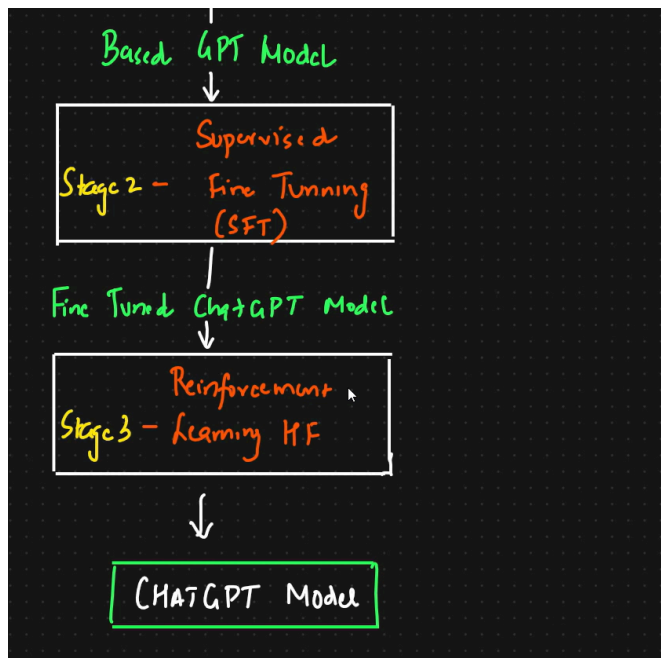
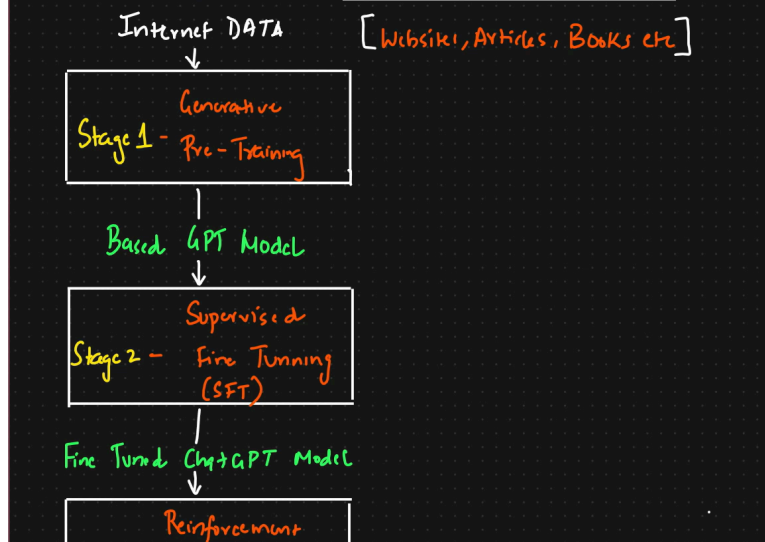




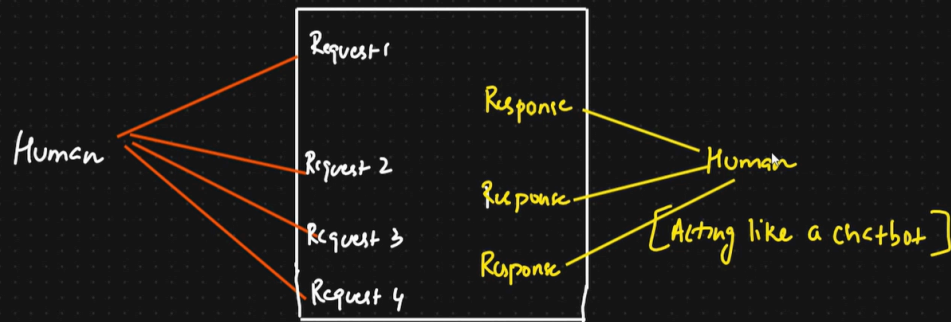
how LLMS are trained

How ChatGPT is TRAINED

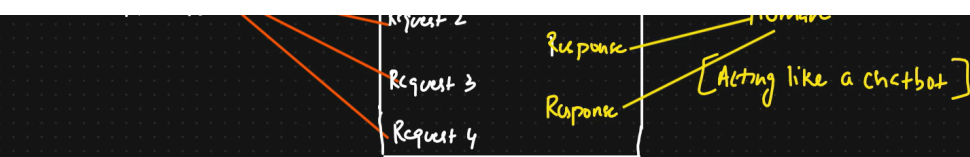


② Supervised Fine Tuning (SFT)

Real Conversation



SFT Training Data Corpus



SFT Training Data Corpus

Request => Conversation history
Response => Best Ideal Response

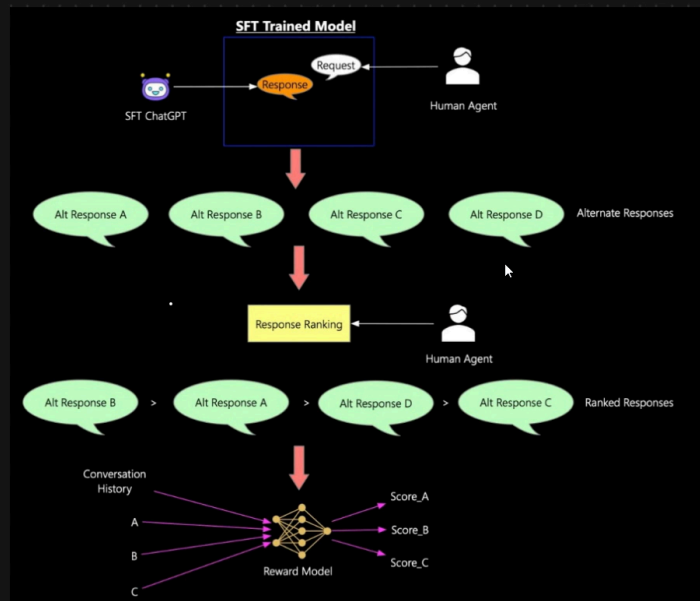
Request Response 1
Request Response 2
Request Response 3

TRAINING
Base GPT Model => SFT Chat GPT Model

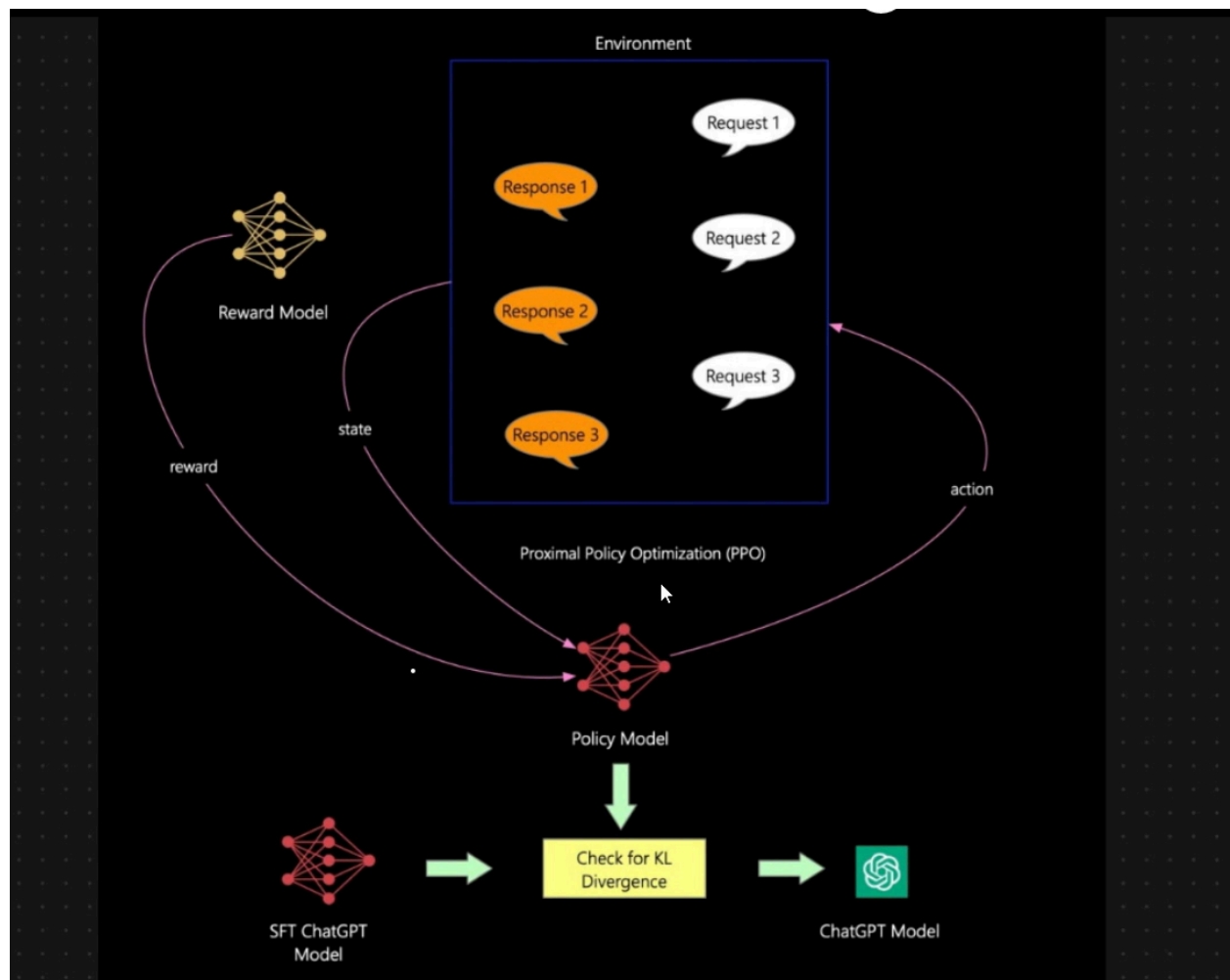
↓
Optimizer
SGD

Reinforcement learning through human feedback (RLHF)

Stage 3: Reinforcement Learning Through Human Feedback (RLHF)



Environment



Langchain EcoSystem

① Why Langchain?
 ② Langchain Ecosystem?

Gen AI App → LLMops
 [Debugging, Evaluation, Monitoring]

↓
 Deploy → API's
 ↓
 AWS

Huggingface Space

lifecycle of Gen AI

Debugging ✓
 Playground ✓
 Evaluation ✓
 Annotation ✓
 Monitoring ✓

Open AI Huggingface Meta Google Gemini Pro

Transformers
 pipelines

IP → LLM → OP

API Key

{open source}

LangChain: Framework to Develop
 Gen AI App.

Hugging Face is a leading open-source platform and community dedicated to natural language processing (NLP) and machine learning, often described as "the GitHub of machine learning." Founded in 2016 by Clément Delangue, Julien Chaumond, and Thomas Wolf, the company originally developed a chatbot app for teenagers before pivoting to provide essential tools for building, sharing, and deploying AI models.

The platform is best known for its Transformers library, which simplifies the use of state-of-the-art pre-trained models for tasks like text generation, translation, and sentiment analysis. It hosts a vast Model Hub containing over 900,000 pre-trained models and 90,000 datasets, allowing developers to browse, download, and fine-tune models for various applications including computer vision and speech recognition.